Subject: DATABASE MANAGEMENT SYSTEMS	**Type:** Assignment Solutions
Student: Kamichetty Veera Pradeepthi (23A91A1225)	**Source:** PEC.UP Smart Academic Portal

Assignment 1 Solutions: SQL Queries & Relational Schema

Question 1: University Database Schema

Write SQL queries to find all students who have enrolled in more than 3 courses in the Academic Year 2025-2026.

Solution:

```
SELECT s.RollNo, s.Name, COUNT(e.CourseID) AS CourseCount

FROM Student s JOIN Enrollment e ON s.RollNo = e.RollNo

WHERE e.AcademicYear = '2025-2026'

GROUP BY s.RollNo, s.Name

HAVING COUNT(e.CourseID) > 3;
```

Question 2: Nested Subqueries & Views

Create a materialized view 'ToppersView' that lists the top 5 students in each branch ranked by SGPA.

Solution:

```
CREATE VIEW ToppersView AS

SELECT RollNo, Name, Branch, SGPA, DENSE_RANK() OVER (PARTITION BY Branch ORDER BY SGPA DESC) as Rank

FROM Student_Semester_Result WHERE Rank <= 5;
```

Downloaded from PEC.UP Academic Resource Hub (https://pecup.in) • Internal Academic Circulation Only